

Gradient descent

$$w \leftarrow w - \eta g$$

Idea 1:

speed from past gradients

Momentum

average of past gradients

NAG

momentum + look ahead

Adam

$$w \leftarrow w - \eta \frac{\hat{m}}{\sqrt{\hat{v}} + \epsilon}$$

$\hat{m}$: average of gradients $\hat{v}$: average of g^2
both corrected for starting at 0

Idea 2:

a learning rate per parameter

AdaGrad

divide by $\sqrt{\text{sum of } g^2}$

RMSProp

divide by $\sqrt{\text{average of } g^2}$